Report on Bangladesh_Agriculture Database Creation

Here's a complete database structure for tracking agricultural data in Bangladesh:

1. Database Creation

```
CREATE DATABASE Bangladesh_Agriculture;
```

2. Crops Table Creation

```
CREATE TABLE Crops (
    crop_id INT PRIMARY KEY,
    crop_name VARCHAR(100) NOT NULL,
    scientific_name VARCHAR(100),
    crop_type VARCHAR(50) NOT NULL,
    growing_season VARCHAR(50),
    water_requirement VARCHAR(50)
);
```

3. Districts Table Creation

```
CREATE TABLE Districts (
    district_id INT PRIMARY KEY,
    district_name VARCHAR(100) NOT NULL,
    division VARCHAR(50) NOT NULL,
    area_sq_km DECIMAL(10,2),
    population INT,
    agricultural_land_percentage DECIMAL(5,2)
);
```

4. Farmers Table Creation

```
CREATE TABLE Farmers (
    farmer_id INT PRIMARY KEY,
    farmer_name VARCHAR(100) NOT NULL,
    district_id INT,
    farm_size DECIMAL(8,2),
    farming_experience INT,
    contact_number VARCHAR(20),
```

```
    FOREIGN KEY (district_id) REFERENCES Districts(district_id)
);
```

5. Production_Data Table Creation

sql

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```
CREATE TABLE Production_Data (
    production_id INT PRIMARY KEY,
    district_id INT,
    crop_id INT,
    year INT NOT NULL,
    production_quantity DECIMAL(12,2),
    cultivated_area DECIMAL(12,2),
    yield_per_hectare DECIMAL(10,2),
    FOREIGN KEY (district_id) REFERENCES Districts(district_id),
    FOREIGN KEY (crop_id) REFERENCES Crops(crop_id)
);
```

6. Agricultural_Inputs Table Creation

```
CREATE TABLE Agricultural_Inputs (
    input_id INT PRIMARY KEY,
    input_name VARCHAR(100) NOT NULL,
    input_type VARCHAR(50) NOT NULL,
    unit VARCHAR(20) NOT NULL,
    avg_price DECIMAL(10,2)
);
```

7. Farmer_Crop_Records Table Creation

```
CREATE TABLE Farmer_Crop_Records (
    record_id INT PRIMARY KEY,
    farmer_id INT,
    crop_id INT,
    season_year VARCHAR(20) NOT NULL,
    cultivated_area DECIMAL(8,2),
    production DECIMAL(10,2),
    input_cost DECIMAL(12,2),
    selling_price DECIMAL(12,2),
    FOREIGN KEY (farmer_id) REFERENCES Farmers(farmer_id),
```

```
    FOREIGN KEY (crop_id) REFERENCES Crops(crop_id)
);
```

8. Market_Prices Table Creation

sql

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```
CREATE TABLE Market_Prices (
    price_id INT PRIMARY KEY,
    crop_id INT,
    district_id INT,
    date DATE NOT NULL,
    wholesale_price DECIMAL(10,2),
    retail_price DECIMAL(10,2),
    FOREIGN KEY (crop_id) REFERENCES Crops(crop_id),
    FOREIGN KEY (district_id) REFERENCES Districts(district_id)
);
```

9. Government_Schemes Table Creation

```
CREATE TABLE Government_Schemes (
    scheme_id INT PRIMARY KEY,
    scheme_name VARCHAR(200) NOT NULL,
    implementing_agency VARCHAR(100),
    start_date DATE,
    end_date DATE,
    budget DECIMAL(15,2),
    target_beneficiaries VARCHAR(200)
);
```

10. Farmer_Scheme_Participation Table Creation

```
CREATE TABLE Farmer_Scheme_Participation (
    participation_id INT PRIMARY KEY,
    farmer_id INT,
    scheme_id INT,
    year INT NOT NULL,
    benefits_received VARCHAR(200),
    amount_received DECIMAL(12,2),
    FOREIGN KEY (farmer_id) REFERENCES Farmers(farmer_id),
    FOREIGN KEY (scheme_id) REFERENCES Government_Schemes(scheme_id));
```

Here are 10 analytical queries with their SQL solutions and expected output formats for the Bangladesh_Agriculture database:

1. Top 5 Most Produced Crops in 2023

Question: Which crops had the highest production quantity in 2023?

```
SELECT c.crop_name, SUM(p.production_quantity) AS total_production
FROM Production_Data p
JOIN Crops c ON p.crop_id = c.crop_id
WHERE p.year = 2023
GROUP BY c.crop_name
ORDER BY total_production DESC
LIMIT 5;
```

Output:

crop_name	total_production
Rice	3850000.50
Potato	1025000.75
Jute	856000.25
Wheat	785000.00
Mango	652000.50

2. Districts with Highest Rice Yield

Question: Which districts have the highest rice yield per hectare?

```
SELECT d.district_name, p.yield_per_hectare
FROM Production_Data p
JOIN Districts d ON p.district_id = d.district_id
JOIN Crops c ON p.crop_id = c.crop_id
WHERE c.crop_name = 'Rice'
ORDER BY p.yield_per_hectare DESC
LIMIT 5;
```

Output:

district_	name yield_per_hectare
Jessore	5200.50
Dinajpur	5150.75
Bogra	5100.25
Naogaon	5050.00
Rangpur	5000.50

3. Farmers with Largest Land Holdings

Question: Who are the top 10 farmers with the largest agricultural land holdings?

```
SELECT farmer_name, farm_size
FROM Farmers
ORDER BY farm_size DESC
LIMIT 10;
```

Output:

farmer_name	farm_size
Abdul	Hamid 85.50
Rahima	Begum 72.25
Mohammad	Ali 65.00
Sufia	Khatun 60.75
Jamal	Uddin 55.50

4. Seasonal Production Comparison

Question: Compare total production between Rabi and Kharif seasons for 2023

```
SELECT c.growing_season, SUM(p.production_quantity) AS total_production
FROM Production_Data p
JOIN Crops c ON p.crop_id = c.crop_id
WHERE p.year = 2023 AND c.growing_season IN ('Rabi', 'Kharif-I', 'Kharif-II')
GROUP BY c.growing_season;
```

Output:

growing_season	total_production
Rabi	4250000.25
Kharif-I	3850000.50
Kharif-II	3200000.75

5. Most Expensive Retail Crops

Question: Which crops had the highest average retail price in 2023?

```
SELECT c.crop_name, AVG(m.retail_price) AS avg_retail_price
FROM Market_Prices m
JOIN Crops c ON m.crop_id = c.crop_id
WHERE YEAR(m.date) = 2023
GROUP BY c.crop_name
ORDER BY avg_retail_price DESC
LIMIT 5;
```

Output:

crop_name	avg_retail_price
Mango	120.50
Litchi	115.75
Pineapple	95.25
Brinjal	45.50
Tomato	40.75

6. Government Scheme Participation

Question: Which government schemes have the highest farmer participation?

```
SELECT g.scheme_name, COUNT(f.participation_id) AS participant_count
FROM Government_Schemes g
JOIN Farmer_Scheme_Participation f ON g.scheme_id = f.scheme_id
GROUP BY g.scheme_name
ORDER BY participant_count DESC
LIMIT 5;
```

Output:

scheme_name	participant_count
Farm Mechanization Program	12500
Seed Subsidy Program	9800
Organic Farming Initiative	8750
Irrigation Support Scheme	7650
Fisheries Development Project	6500

7. Input Cost vs Selling Price Analysis

Question: Show the average input cost and selling price for each crop type

```
SELECT c.crop_type,
       AVG(f.input_cost) AS avg_input_cost,
       AVG(f.selling_price) AS avg_selling_price
FROM Farmer_Crop_Records f
JOIN Crops c ON f.crop_id = c.crop_id
GROUP BY c.crop_type;
```

Output:

crop_type	avg_input_cost	avg_selling_price
Cereal	12500.50	18500.75
Vegetables	8500.25	12000.50
Fruits	15000.75	25000.25
Pulses	9500.00	13500.50

8. District-wise Agricultural Land

Question: Which districts have the highest percentage of agricultural land?

```
SELECT district_name, agricultural_land_percentage
FROM Districts
ORDER BY agricultural_land_percentage DESC
LIMIT 10;
```

Output:

district_name	agricultural_land_percentage
Kushtia	85.75
Pabna	82.50
Bogra	80.25
Rangpur	78.50
Dinajpur	77.75

9. Year-over-Year Production Growth

Question: Show the year-over-year production growth for rice

```
SELECT year, production_quantity,
       LAG(production_quantity) OVER (ORDER BY year) AS prev_year,
       ROUND(((production_quantity - LAG(production_quantity) OVER (ORDER BY year)) /
       LAG(production_quantity) OVER (ORDER BY year)) * 100, 2) AS growth_percentage
FROM Production_Data p
JOIN Crops c ON p.crop_id = c.crop_id
WHERE c.crop_name = 'Rice'
ORDER BY year;
```

Output:

year	production_quantity	prev_year	growth_percentage
2020	3500000.25	NULL	NULL
2021	3650000.50	3500000.25	4.29
2022	3750000.75	3650000.50	2.74
2023	3850000.50	3750000.75	2.67

10. Most Profitable Crops

Question: Which crops provide the highest profit margin (selling price vs input cost)?

```
SELECT c.crop_name,
       AVG(f.selling_price - f.input_cost) AS avg_profit,
       ROUND(AVG((f.selling_price - f.input_cost) / f.input_cost * 100, 2) AS profit_
margin
FROM Farmer_Crop_Records f
JOIN Crops c ON f.crop_id = c.crop_id
```

```
GROUP BY c.crop_name
ORDER BY avg_profit DESC
LIMIT 5;
```

Output:

CROP_NAME	AVG_PROFIT	PROFIT_MARGIN
MANGO	12500.50	85.25
LITCHI	11500.75	80.50
PINEAPPLE	9500.25	75.75
TOMATO	6500.50	70.25
BRINJAL	6000.75	68.50